Yao-Qi Zhang

Curriculum Vitae

Postdoctoral Researcher, University of Hertfordshire

y.zhang59@herts.ac.uk | INSPIRE-HEP | ORCID 0000-0001-9211-1952

Research interests

Quantum field theory, string theory, and mathematical physics, with a focus on scattering amplitudes, positive geometries, cosmological wavefunctions, correlators, and their physical applications.

Appointments and education

University of Hertfordshire 2026–present
Postdoctoral Researcher

Institute of Theoretical Physics, Chinese Academy of Sciences 2020–2025
Ph.D. Student in Theoretical Physics; advisor: Song He

Beijing Normal University 2016–2020
B.S. in Physics

Presentations

- "All-loop ABJM amplitudes from projected, bipartite amplituhedron," Workshop on Geometry and Scattering Amplitudes, August 2022.

Publications

1. **de Sitter Wavefunction from Quadrangular Polylogarithms: Polygon Graphs**
   Livia Ferro, Tomasz Lukowski, Shruti Paranjape, Lecheng Ren, Marcus Spradlin, Anastasia Volovich, He-Chen Weng, and Yao-Qi Zhang.
   Preprint (2026) [arXiv:2610.06772]
2. **de Sitter Wavefunction from Quadrangular Polylogarithms: Chain Graphs**
   Livia Ferro, Tomasz Lukowski, Lecheng Ren, Marcus Spradlin, Anastasia Volovich, He-Chen Weng, and Yao-Qi Zhang.
   Preprint (2026) [arXiv:2605.06542] [INSPIRE]
3. **Generalised Cluster Adjacency for Cosmology**
   Mattia Capuano, Livia Ferro, Tomasz Lukowski, Alessandro Palazio, and Yao-Qi Zhang.
   Preprint (2026) [arXiv:2603.09965] [INSPIRE]
4. **Loops and legs: ABJM amplitudes from f-graphs**
   Song He and Yao-Qi Zhang.
   JHEP 05 (2026) 218. [arXiv:2601.21980] [DOI] [INSPIRE]
5. **Hidden Permutation Symmetry of Squared Amplitudes in Aharony-Bergman-Jafferis-Maldacena Theory**
   Song He, Canxin Shi, Yichao Tang, and Yao-Qi Zhang.
   Phys. Rev. Lett. 135 (2025) 211601. [arXiv:2508.03813] [DOI] [INSPIRE]
6. **Notes on conformal integrals: Coulomb branch amplitudes, magic identities, and bootstrap**
   Song He, Xuhang Jiang, Jiahao Liu, and Yao-Qi Zhang.
   Phys. Rev. D 112 (2025) 076012. [arXiv:2502.08871] [DOI] [INSPIRE]
7. **Differential equations and recursive solutions for cosmological amplitudes**
   Song He, Xuhang Jiang, Jiahao Liu, Qinglin Yang, and Yao-Qi Zhang.
   JHEP 01 (2025) 001. [arXiv:2407.17715] [DOI] [INSPIRE]

8. **The cusp limit of correlators and a new graphical bootstrap for correlators/amplitudes to eleven loops**
Song He, Canxin Shi, Yichao Tang, and Yao-Qi Zhang.
JHEP 03 (2025) 192. [arXiv:2410.09859] [DOI] [INSPIRE]

9. **From squared amplitudes to energy correlators**
Song He, Xuhang Jiang, Qinglin Yang, and Yao-Qi Zhang.
Preprint (2024). [arXiv:2408.04222] [INSPIRE]

10. **One-loop Bern-Carrasco-Johansson numerators on quadratic propagators from the worldsheet**
Jin Dong, Yao-Qi Zhang, and Yong Zhang.
Phys. Rev. D 109 (2024) L101905. [arXiv:2312.01580] [DOI] [INSPIRE]

11. **Emergent unitarity, all-loop cuts and integrations from the ABJM amplituhedron**
Song He, Chia-Kai Kuo, Zhenjie Li, and Yao-Qi Zhang.
JHEP 07 (2023) 212. [arXiv:2303.03035] [DOI] [INSPIRE]

12. **Covariant color-kinematics duality, Hopf algebras, and permutohedra**
Qu Cao, Jin Dong, Song He, and Yao-Qi Zhang.
Phys. Rev. D 107 (2023) 026022. [arXiv:2211.05404] [DOI] [INSPIRE]

13. **One-loop diagrams with quadratic propagators from the worldsheet**
Bo Feng, Song He, Yong Zhang, and Yao-Qi Zhang.
JHEP 08 (2022) 240. [arXiv:2204.13659] [DOI] [INSPIRE]

14. **All-Loop Four-Point Aharony-Bergman-Jafferis-Maldacena Amplitudes from Dimensional Reduction of the Amplituhedron**
Song He, Chia-Kai Kuo, Zhenjie Li, and Yao-Qi Zhang.
Phys. Rev. Lett. 129 (2022) 221604. [arXiv:2204.08297] [DOI] [INSPIRE]

15. **The momentum amplituhedron of SYM and ABJM from twistor-string maps**
Song He, Chia-Kai Kuo, and Yao-Qi Zhang.
JHEP 02 (2022) 148. [arXiv:2111.02576] [DOI] [INSPIRE]

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Polygon Graphs added from arXiv:2610.06772 on 13 October 2026.